KITCHEN SINK

one of everything, with the Markdown that draws it in the README

1 TEXT.

Plain Markdown paragraphs justify in the Roman face. **Bold**, *italic*, underline, `typewriter`, H_2O and 10^6 all work inline, and a spaced double hyphen – like this – prints as an en dash. Links keep their words and drop the URL, because paper has no links: see the README.

Inline math sits in the line: the time constant is $\tau = RC$ and the corner is $f_c = 1/(2\pi RC)$.

$$v(t) = V_0\, e^{-t/\tau} \tag{1-1}$$

1.1 Lists and code.

- an unordered list item
- a second one, which wraps onto a following line when it runs long enough to need more than one line of the sheet
- a third

1. an ordered list
2. numbers itself

```
V16 N68 E      monitor range, altitude rate, altitude
V57 E          landing radar update permitted
```

---

2 TABLES.

A bare pipe table renders ruled and untitled. Wrap it in a table block to number and title it.

Mode	Gain	Rate
coarse	4.0	2.0 deg/s
fine	0.5	0.2 deg/s

Table 1-1. Thruster Quad Assignments

Quad	Axis	Thrusters	Min impulse
A	pitch, roll	A1U A2D A3A A4F	0.4 N s
B	yaw, roll	B1L B2R B3A B4F	0.4 N s
C	pitch, roll	C1U C2D C3A C4F	0.4 N s
D	yaw, roll	D1L D2R D3A D4F	0.4 N s

3 CALL-OUTS.

NOTE

A note is advice: nothing breaks if it is ignored.

CAUTION

A caution protects the equipment. Do not command fine mode with a quad isolated.

WARNING

A warning protects the crew. It is boxed so it cannot be missed.

3.1 Change bars.

Anything inside a changed block gets a black bar in the left gutter, the handbook's way of saying "this paragraph is new since the last issue".

QUAD ISOLATION	CDR
QUAD A ISOL VLV	CLOSE
verify talkback	BARBERPOLE
RCS SYS A/B	CHECK PRESS
ATT CONTROL	MODE CONT

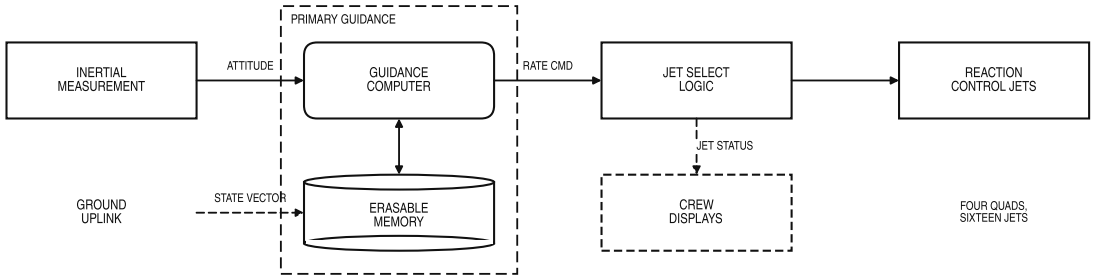
PART I

FIGURES

I.1	Block diagrams
I.2	Graphs
I.3	Timelines
I.4	Plots
I.5	Panels
I.6	Stacks and bit fields
I.7	Bars and matrices
I.8	Points and spreads
I.9	Flows
I.10	Images and callouts
I.11	Trees
I.12	Shares and choices
I.13	Totals and changes
I.14	Fields, directions and many measures
I.15	Networks, lanes and plans
I.16	Maps and sparklines

I.1 BLOCK DIAGRAMS.

Nodes sit on a grid of columns and rows. Shapes: box, round, drum (storage), dash (outside the system) and text (lettering only). A fence draws a dashed boundary round a subsystem; edges may be dashed, two-headed and labelled.

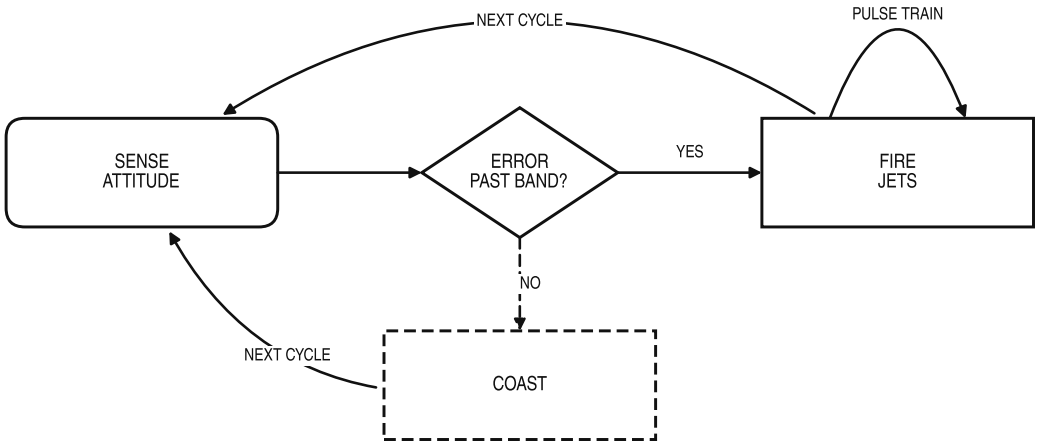


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Figure 1-1. Attitude Control Functional Diagram, Every Shape and Edge

I.2 GRAPHS.

A diagram is also a graph: a diamond is a decision, `bend=` curves a back-edge, and a loop returns a node to itself.

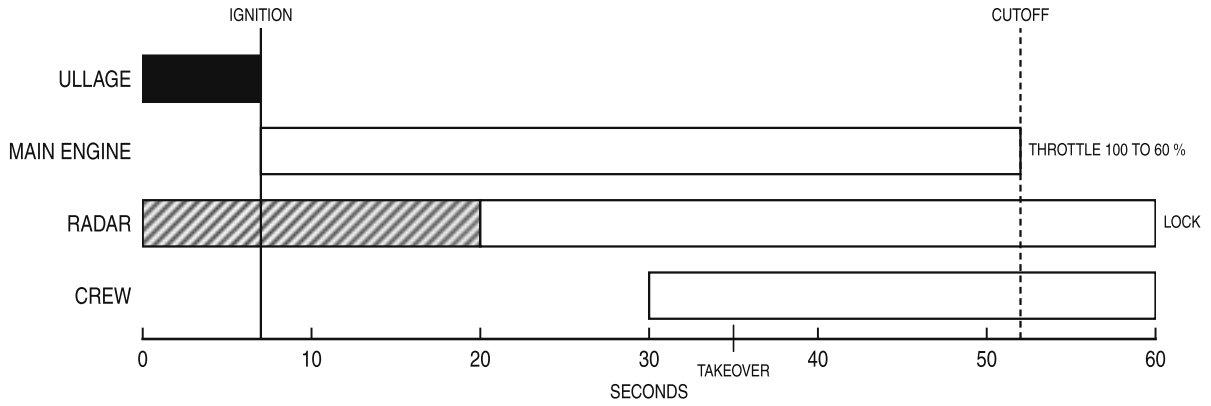


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Figure 1-2. Attitude Hold, a Decision Loop

I.3 TIMELINES.

Bars may be hatched, open or solid; marks are vertical lines with a label above, and a call drops a leader from a bar to lettering below the axis.



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Figure 1-3. Sequence of Events, Every Bar and Mark

I.4 PLOTS.

Lines come from a CSV and are labelled at their right ends instead of with a legend; marks draw a vertical line. Log axes are a single line.

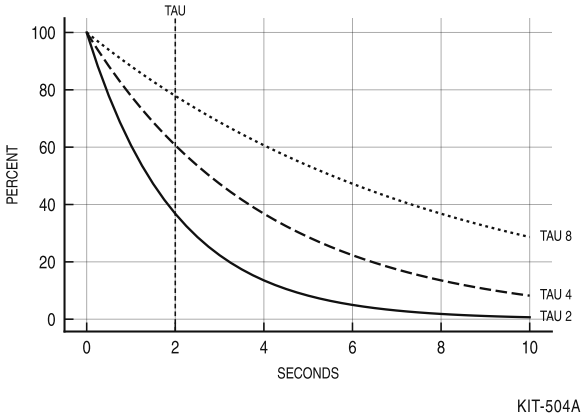


Figure 1-4. Decay of Three Time Constants

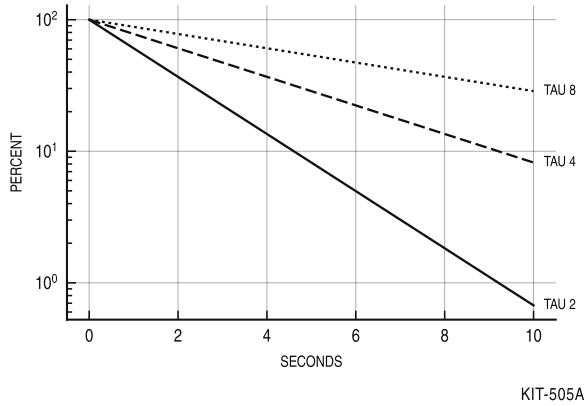


Figure 1-5. The Same, on a Log Axis

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I.5 PANELS.

A panel line starts a second panel under the first, on the same x; fn takes degrees() and radians().

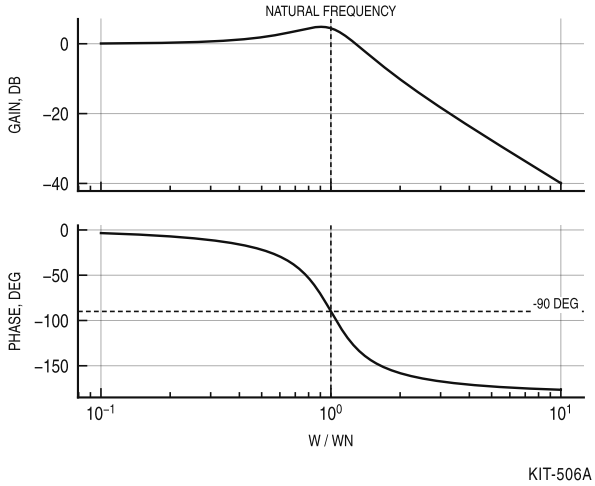


Figure 1-6. Rate Loop, Gain and Phase

I.6 STACKS AND BIT FIELDS.

A stack sets layers top to bottom; a bit field wraps fields into rows of bits, hatching the reserved ones.

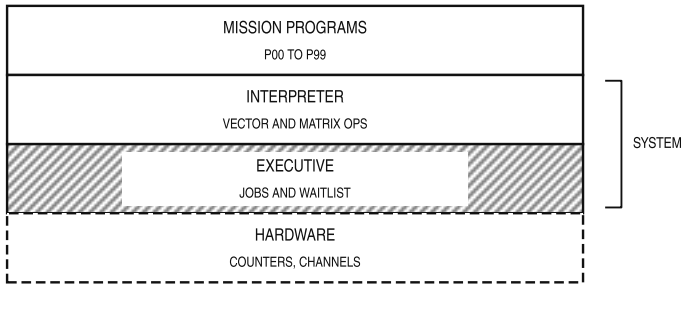


Figure 1-7. Flight Software Layers

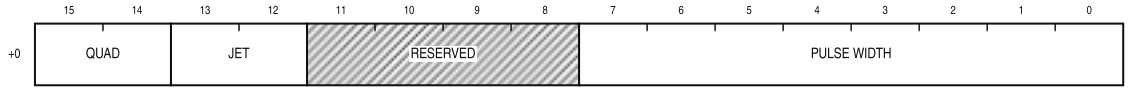


Figure 1-8. Jet Command Word

I.7 BARS AND MATRICES.

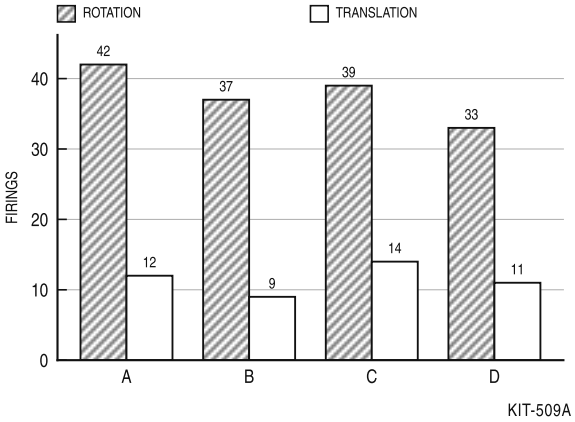


Figure 1-9. Firings by Quad

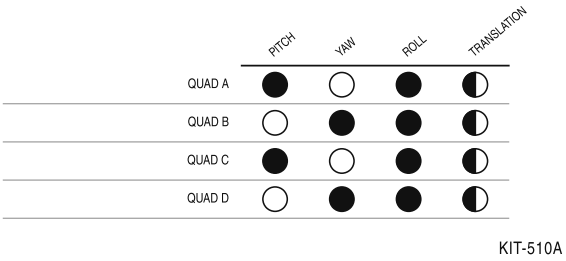


Figure 1-10. Quad Authority

With shade, a matrix of numbers is a heatmap: each cell tinted by its value.

	ULLAGE	COAST	BRAKING	APPROACH	LANDING
QUAD A	4	2	9	14	21
QUAD B	4	1	7	12	18
QUAD C	4	2	8	15	20
QUAD D	4	1	6	11	17

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Figure 1-11. Jet Firings by Quad and Phase

I.8 POINTS AND SPREADS.

A scatter letters its points and fits a line; a dist draws boxes from five numbers (or raw CSV values).

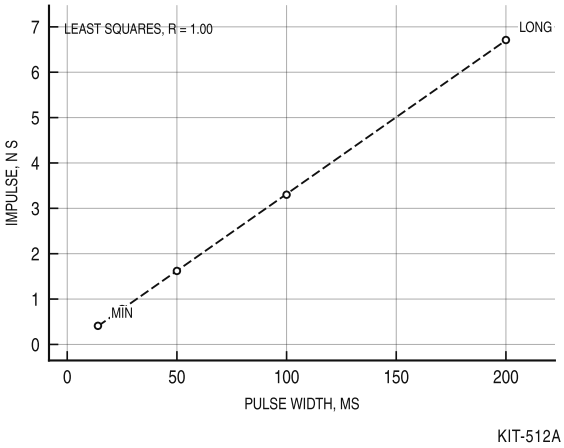


Figure 1-12. Pulse Width and Impulse

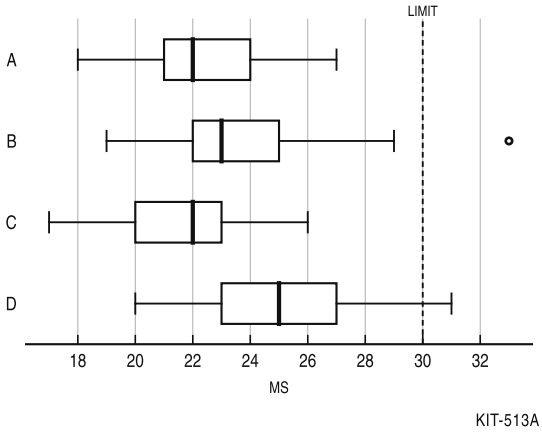


Figure 1-13. Jet Response by Quad

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I.9 FLOWS.

A sankey draws flows between stages as ribbons as wide as their values.

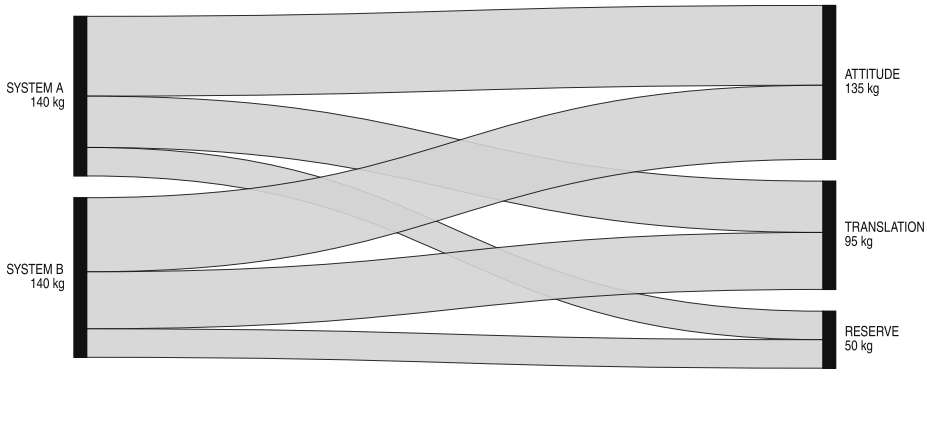


Figure 1-14. Propellant Budget, kg

I.10 IMAGES AND CALLOUTS.

Any SVG or PNG. Width may be half or third; gray=1 prints a dark screen capture as ink on paper. A callouts block draws leaders from points on a picture to lettering in its margins, on the nearer side.

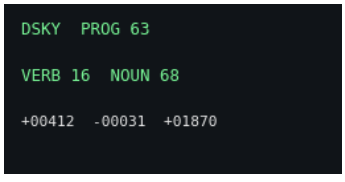


Figure 1-15. A screen capture, as captured

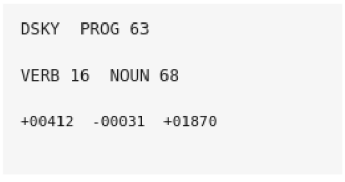
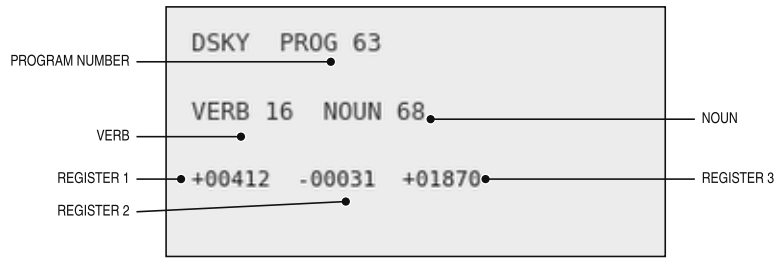


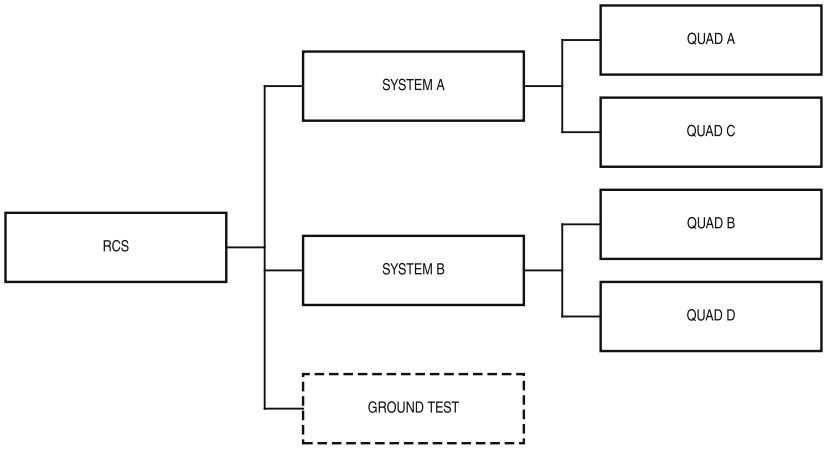
Figure 1-16. The same capture with gray=1



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Figure 1-17. The Display, Named

I.11 TREES.

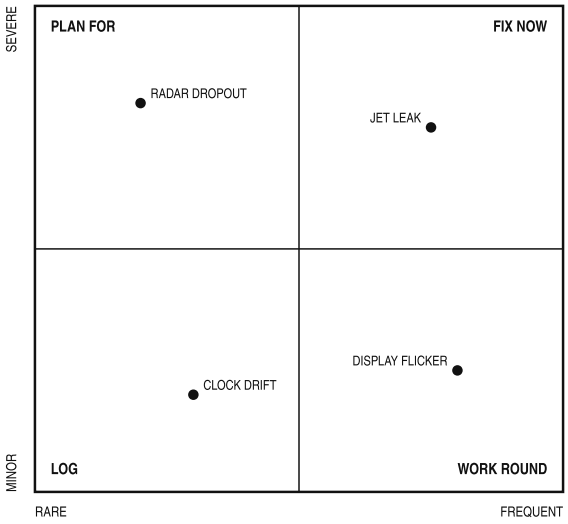


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Figure 1-18. Reaction Control, by System

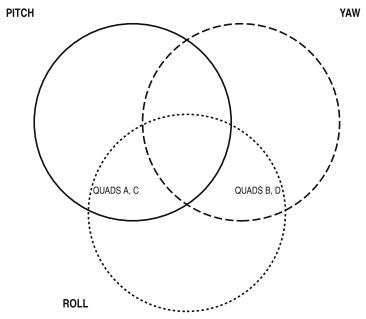
I.12 SHARES AND CHOICES.

A quadrant places items on two judgements; a Venn letters overlaps; a treemap nests shares; a funnel narrows.



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Figure 1-19. Anomalies to Work



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Figure 1-20. Jets by Duty

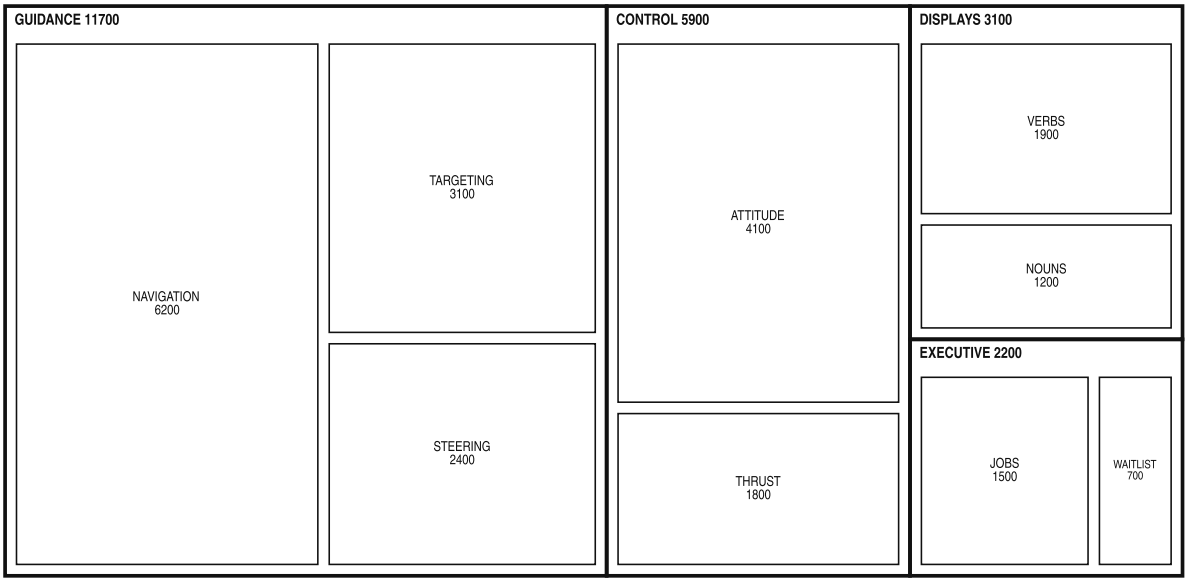


Figure 1-21. Flight Software by Module, Words

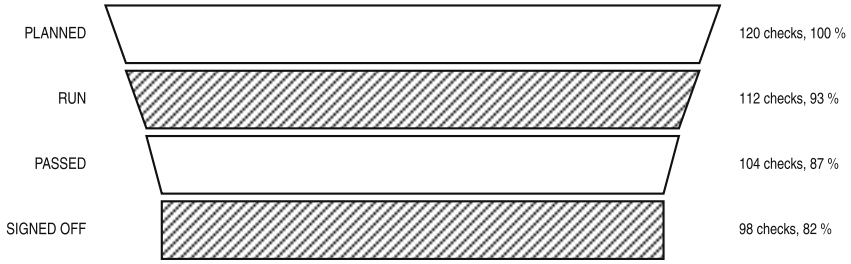


Figure 1-22. Pre-Flight Checks

I.13 TOTALS AND CHANGES.

A waterfall walks a running total; a slope runs each thing from one build to the next; a waffle counts shares.

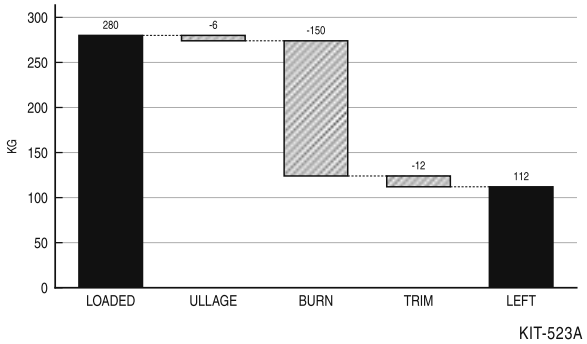


Figure 1-23. Propellant over a Burn, kg

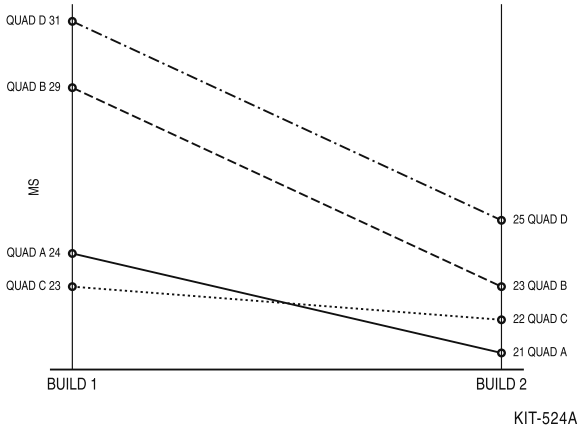


Figure 1-24. Jet Response by Build, ms

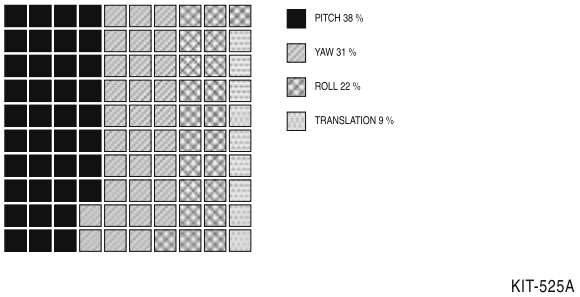


Figure 1-25. Thruster Time by Axis

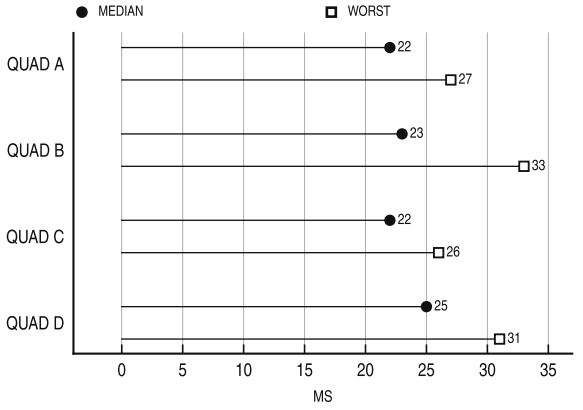


Figure 1-26. Jet Response, Median and Worst

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I.14 FIELDS, DIRECTIONS AND MANY MEASURES.

A contour draws a field with its arrows; a raster sets it as gray; a polar plot draws a pattern or a wind rose; parallel axes give each measure its own axis.

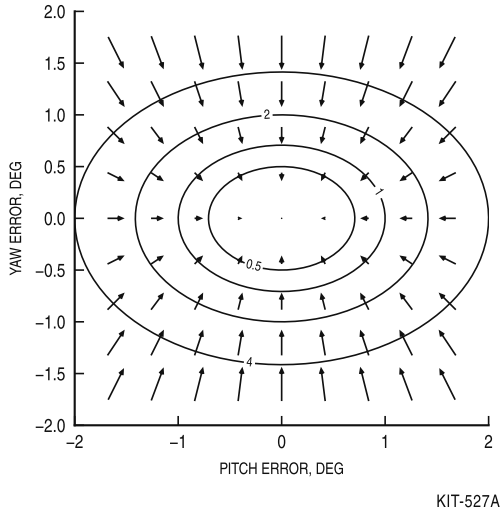


Figure 1-27. Attitude Error Potential

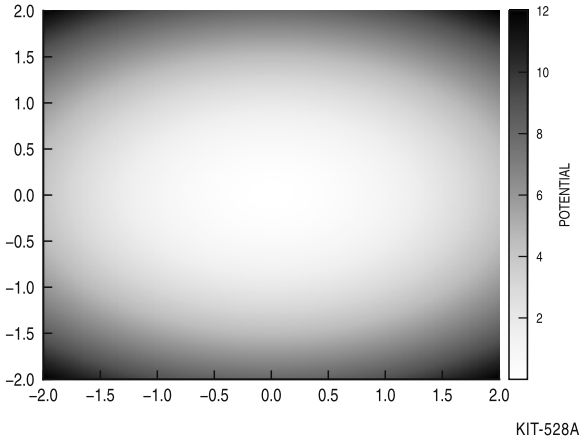


Figure 1-28. The Same, as Gray

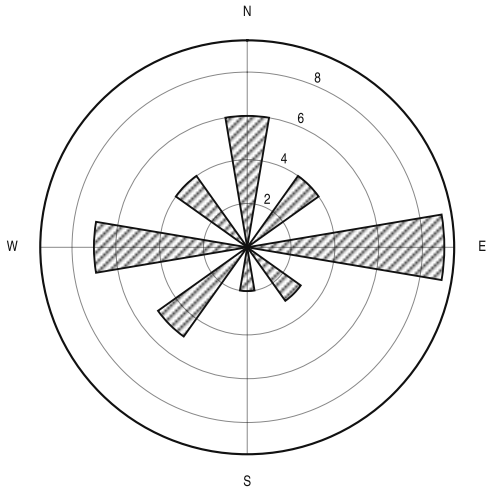


Figure 1-29. Landing-Site Winds, Example

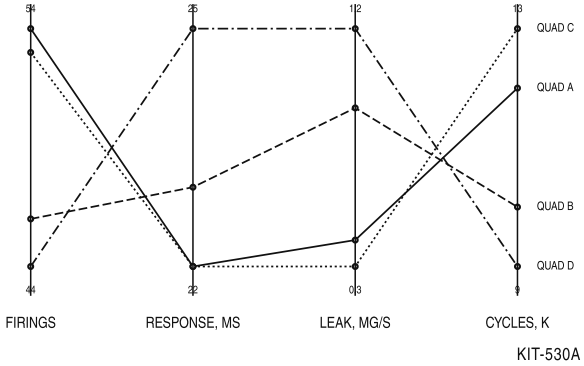


Figure 1-30. Four Quads, Four Measures

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I.15 NETWORKS, LANES AND PLANS.

A network lays out a graph for you; lanes give each row of a diagram an owner; milestones and dependencies make a timeline a plan.

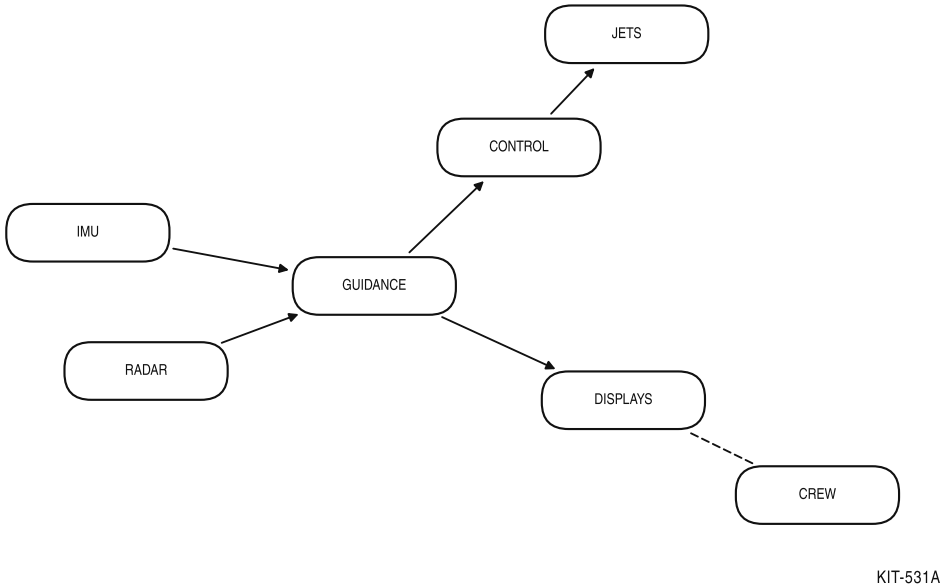


Figure 1-31. Who Talks to Whom

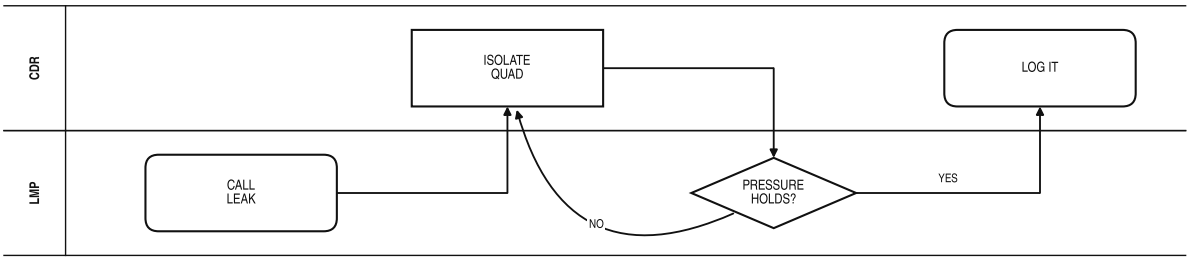


Figure 1-32. Quad Isolation, by Crew Member

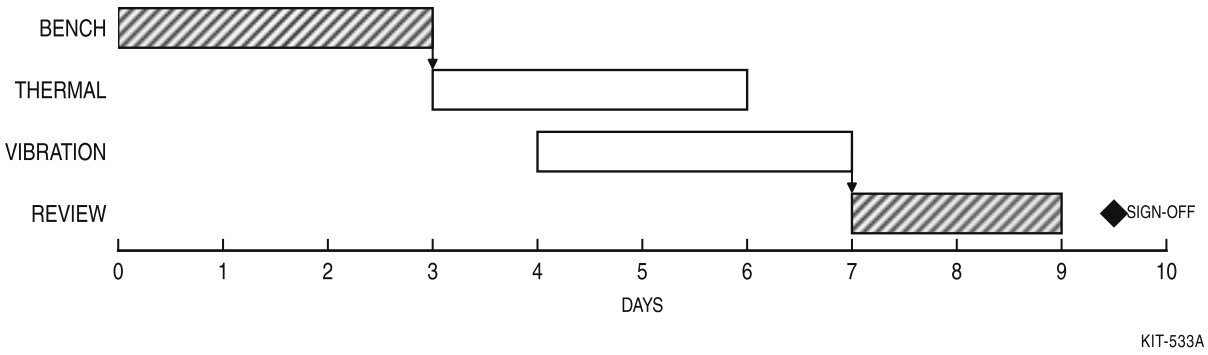


Figure 1-33. Bring-Up Plan, Days

I.16 MAPS AND SPARKLINES.

A map draws GeoJSON or shapefile layers with points, routes, a graticule and a scale. A sparkline sits in a line of text, the last value dotted: quad B's response over six builds ran , ms.

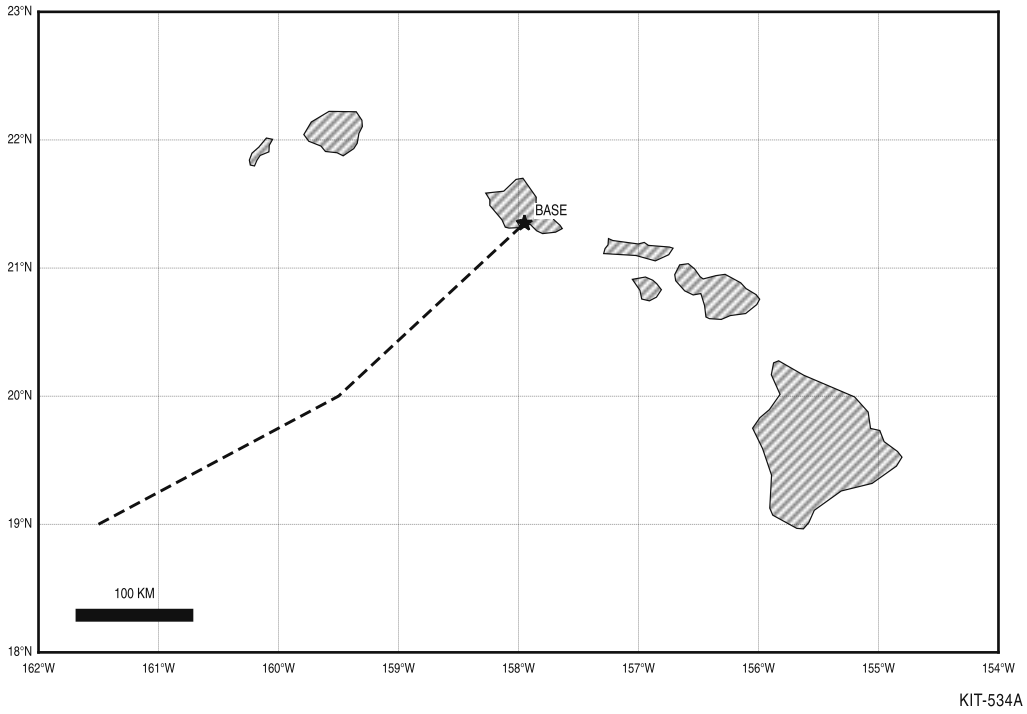


Figure 1-34. Recovery Area, Example